INCH-POUND

MIL-T-27/296B
20 April 1993
SUPERSEDING
MIL-T-27/296A
4 November 1980

MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, TF5V03ZZ

(B) Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.

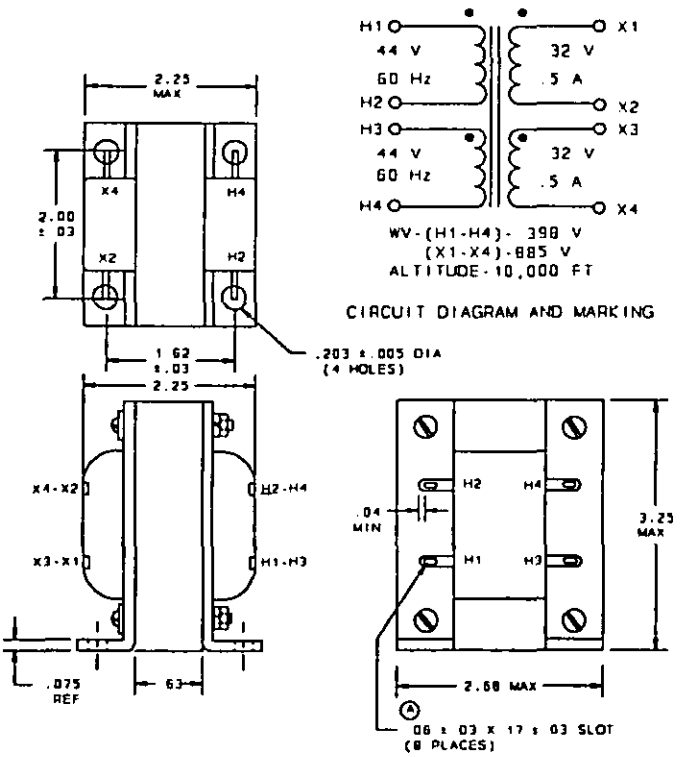


FIGURE 1. Dimensions and configurations.

(B) denotes changes

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Inches	mm
.005	0.13
.03	0.8
.04	1.0
.06	1.5
.075	1.90
.17	4.3
.203	5.16
.63	16.0
1.62	41.1
2.00	50.8
2.25	57.2
2.68	68.1
3.25	82.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Marking shall be on the sides of the case.
4. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three-place decimals and ±.03 (0.8 mm) for two-place decimals.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (H1-H2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary voltage (H1-H4): 88 V rms, 60 Hz.

Primary current (H1-H4): .5 ampere rms.

Volt ampere: 22 voltamperes.

Secondary:

(X1-X2): 32 volts, .5 ampere rms.
(X3-X4): 32 volts, .5 ampere rms.

Working voltage:

(H1-H4): 389 volts, maximum.
(X1-X4): 885 volts, maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder type.

Material: Brass, tinned or cadmium plated.

Height: .25 inch, minimum.

Weight: 28.6 ounces, maximum.

Operating temperature range: +5°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 2,240 volts rms, except H1 to H3 shall be 1,090 volts rms.

Electrical characteristics:

No load: With 44 volts, 60 Hz across (H1-H2) or (H3-H4), current shall not exceed 100 milliamperes and the power shall not exceed 8.0 watts.

Voltage across (X1-X2) and (X3-X4): 43 volts ±1 percent.

Rated load: With 44 volts, 60 Hz across (H1-H2) and (H3-H4), and a .5 ampere resistive load in (X1-X2) and (X3-X4) the voltage across (X1-X2) and (X3-X4) shall be 32 volts ±5 percent.

DC resistance:

(H1-H2), (H3-H4): 8.7 ohms ±20 percent.
(X1-X2), (X3-X4): 13.6 ohm ±20 percent.

Polarity: Additive, with terminals H2 and H3 connected, or terminals H2 and X1 or terminals H2 and X3.

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Temperature rise: 90°C maximum with 88 volts, rms, 60 Hz across (H1-H4) with H2 and H3 connected.

Shock (high impact): MIL-STD-202, method 207.

(B) Quality assurance provisions:

Qualification inspection: Not applicable for this specification sheet.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/296-01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army - AR, WC
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0815-31)

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